

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 14

ADDITION OF TWO 16 – BIT NUMBERS

AIM: To write and execute an 8085 assembly language program to add two 16 – bit numbers and store the result in memory.

ALGORITHM

1. Load the first 16-bit number into register AX.
2. Load the second 16-bit number into register BX.
3. Add the contents of BX to AX.
4. Store the result in memory.
5. Stop the program.

CODE:

```
MOV AX, 1234H
MOV BX, 2345H
ADD AX, BX
MOV [8050H], AX
HLT
```

OUTPUT:

The screenshot displays the emu8086 - assembler and microprocessor emulator 4.08 interface. The main window shows the assembly code for a program named 'noname.bin_'. The code is as follows:

```
01 MOV AX, 1234H
02 MOV BX, 2345H
03 ADD AX, BX
04
05 MOV [0050H], AX
06
07 HLT
```

The registers window shows the following values:

Register	H	L
AX	35	79
BX	23	45
CX	00	00
DX	00	00
CS	0100	
IP	0000	
SS	0100	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0100	
ES	0100	

The memory window shows the following values:

Address	Value
01000	B8 184 3
01001	34 052 4
01002	12 018 5
01003	BB 187 6
01004	45 069 7
01005	23 035 8
01006	03 003 9
01007	C3 195 A
01008	A3 163 B
01009	50 080 C
0100A	80 128 D
0100B	F4 244 E
0100C	90 144 F
0100D	90 144 10
0100E	90 144 11
0100F	90 144 12
01010	90 144 13
01011	90 144 14
01012	90 144 15
01013	90 144 16
01014	90 144 17
01015	90 144 18

The original source code window shows the following code:

```
01 MOV AX, 1234H
02 MOV BX, 2345H
03 ADD AX, BX
04
05 MOV [0050H], AX
06
07 HLT
```